

CARMi, IL, USA

WMO: 722074

Lat: 38.089N

Lon: 88.123W

Elev: 118

StdP: 99.91

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 63840

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-12.7	-10.1	-17.5	0.8	-11.6	-15.0	1.0	-9.1	12.5	5.6	11.5	2.6	2.9	320	0.459

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.9	34.3	24.3	32.9	24.1	32.1	23.9	26.8	31.1	25.8	30.6	25.1	30.0	3.6	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.6	21.1	29.5	24.4	19.7	28.4	23.6	18.7	27.5	84.2	31.1	80.2	30.8	76.9	30.2	29.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.5	8.9	7.9	DB	-16.8	36.5	2.2	1.8	-18.5	37.8	-19.8	38.9	-21.0	39.9	-22.6	41.3	(4)
(5)			WB	-17.4	27.9	2.1	1.2	-18.9	28.8	-20.1	29.5	-21.2	30.1	-22.7	31.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	13.7	0.7	2.5	7.7	14.2	19.6	24.5	24.8	24.2	20.9	14.2	7.5	3.0
	DBStd	9.96	6.14	6.15	5.60	5.10	4.56	2.93	2.56	2.86	3.65	5.15	5.17	5.31
	HDD10.0	991	292	221	111	19	1	0	0	0	0	16	107	224
	HDD18.3	2472	546	444	329	141	42	1	0	1	16	149	327	474
	CDD10.0	2346	4	10	41	144	299	436	458	439	328	147	31	8
	CDD18.3	784	0	0	2	16	82	187	199	182	94	22	1	0
	CDH23.3	7658	0	1	16	161	752	1911	1893	1698	971	251	6	0
	CDH26.7	3068	0	0	1	26	257	837	781	703	390	72	1	0
Wind	WSAvg	3.2	4.1	4.0	4.1	4.0	3.3	2.7	2.0	1.7	2.0	3.0	3.4	3.6
Precipitation	PrecAvg	1160	91	71	99	114	134	100	92	87	82	85	105	91
	PrecMax	1424	194	149	226	305	268	207	189	236	163	217	215	177
	PrecMin	898	20	26	47	44	47	41	31	10	4	14	7	31
	PrecStd	165	39	33	46	65	69	42	46	60	48	53	48	38
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	17.7	21.4	25.4	29.8	33.0	37.1	36.2	36.1	34.9	32.0	23.8	19.0
		MCWB	14.2	15.7	17.2	18.8	22.4	22.5	24.5	24.5	22.7	21.2	16.9	16.3
	2%	DB	15.8	18.3	22.4	27.2	31.1	34.0	33.5	33.8	32.7	28.5	21.2	16.2
		MCWB	13.2	14.7	15.7	18.3	21.5	24.0	25.4	24.7	23.2	20.1	15.8	14.4
	5%	DB	12.7	15.2	19.8	25.2	29.7	32.7	32.3	32.3	31.1	26.3	18.4	13.6
		MCWB	10.6	12.1	14.4	17.2	20.8	23.4	25.2	24.5	22.8	18.8	14.1	11.5
	10%	DB	9.9	12.4	17.0	23.0	27.8	31.4	31.0	30.9	28.9	23.3	16.2	11.2
		MCWB	8.0	9.3	12.9	16.6	20.2	23.1	24.5	24.3	21.4	17.3	12.1	9.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	15.8	17.2	18.4	21.2	24.0	26.6	28.1	27.9	25.6	23.4	19.5	16.9
		MCDB	17.3	19.6	23.1	26.6	29.2	31.8	32.4	31.9	31.1	29.1	22.5	18.6
	2%	WB	13.5	15.1	16.8	19.9	23.0	25.6	27.0	26.6	24.7	21.8	17.1	14.7
		MCDB	15.1	17.7	21.2	25.1	28.7	31.5	31.2	30.9	30.4	26.2	19.5	16.1
	5%	WB	11.1	12.7	15.2	18.6	22.2	24.9	26.0	25.6	23.7	20.1	14.7	12.0
		MCDB	12.5	14.8	18.4	23.6	27.5	30.7	30.3	30.4	29.1	24.4	17.6	13.0
	10%	WB	7.7	9.4	13.4	17.5	21.2	24.1	25.1	24.8	22.6	18.5	12.6	9.3
		MCDB	9.4	11.8	16.7	21.3	26.1	29.2	29.5	29.3	27.0	22.3	15.7	10.9
Mean Daily Temperature Range	5% DB	MDBR	8.5	9.3	10.2	11.9	11.2	11.5	10.9	11.7	13.4	12.9	11.0	8.3
		MCDBR	12.1	13.7	14.0	14.3	13.1	13.1	12.2	13.5	15.5	15.7	14.4	11.4
		MCWBR	10.1	10.5	8.6	7.4	5.3	4.9	5.5	5.9	6.6	7.8	9.4	9.4
	5% WB	MCDBR	10.9	12.6	12.6	11.9	11.5	11.4	10.8	11.3	13.2	13.2	12.1	10.2
		MCWBR	10.1	10.5	8.7	7.3	5.3	5.0	5.7	5.8	6.2	7.0	9.6	9.4
Clear-Sky Solar Irradiance	taub		0.314	0.327	0.366	0.403	0.437	0.445	0.472	0.475	0.413	0.361	0.326	0.311
	taud		2.456	2.399	2.350	2.254	2.212	2.210	2.145	2.125	2.291	2.431	2.472	2.468
	Ebn at Noon		867	900	893	876	848	839	812	800	833	850	845	844
	Edh at Noon		82	99	115	134	142	143	151	151	121	95	81	76
All-Sky Solar Radiation	RadAvg		1.99	2.72	3.75	4.95	5.57	6.38	6.16	5.63	4.69	3.45	2.34	1.71
	RadStd		0.20	0.31	0.35	0.29	0.48	0.40	0.39	0.30	0.32	0.36	0.26	0.15

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (35 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page